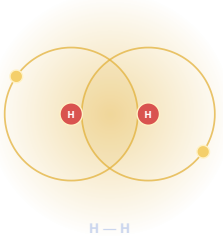
MT Classes, Motihari

Atomic Foundations of Matter



Class 9 Science, Chapter 9 (NCERT Exploration, new syllabus)

Laws of chemical combination, Dalton's theory, covalent and ionic bonds, formulae and molecular mass, with 25 textbook diagrams, every NCERT question solved, and a quick quiz.

Start Think it over

In Chapter 8 (**Journey Inside the Atom**) you learnt that atoms with 8 electrons in their outermost shell (an **octet**) are stable. Atoms that do not have an octet **lose, gain or share** electrons to get one. This chapter shows what happens when they do: atoms join to make molecules and compounds, and the mass never changes along the way.

Example to keep in mind

Hydrogen (a gas that burns) + oxygen (a gas that helps burning) → **water** (a liquid that puts out fire). The properties change completely, but the **mass of water formed** = **mass of hydrogen** + **mass of oxygen** that combined.

Questions from the chapter opening

Try to answer first, then tap to check.

– **Water can be obtained from various sources. Are all these samples of water chemically identical?**

Yes. Pure water from a river, a borewell or the sea always contains hydrogen and oxygen in the mass ratio **1 : 8**. This is the **Law of Constant Proportions** (Section 9.2). Only the dissolved impurities differ.

– **Oxygen is sometimes written as O and sometimes as O_2. What is the difference?**

O means one **atom** of oxygen. **O_2** means one **molecule** of oxygen, made of two oxygen atoms joined by a double covalent bond (O=O). Oxygen gas in the air exists as O_2 molecules.

– **Why does dissolved salt in water conduct electricity, but sugar does not?**

Salt (NaCl) is an **ionic** compound. In water its Na^+ and Cl^- ions become free to move and carry current. Sugar is a **covalent** compound. It dissolves as whole molecules and gives no ions, so its solution does not conduct (Section 9.6).

9.1 Law of Conservation of Mass

Activity 9.1: Does mass change in a physical change?

1. Put a dry 100 mL beaker on a digital balance and press **tare** (reset to zero).
2. Pour about 50 mL of water and add a spatula of common salt. Note the reading.
3. Swirl until the salt dissolves and note the reading again.

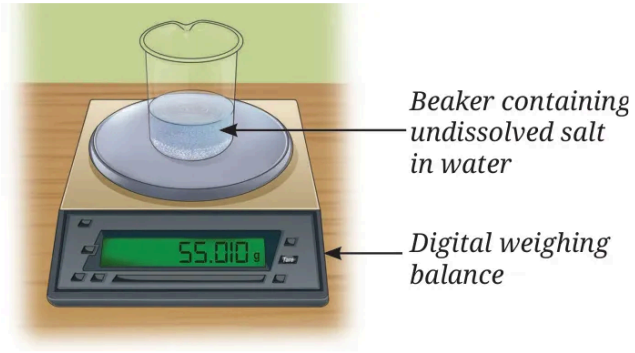


Fig. 9.1 (a) Water + undissolved salt

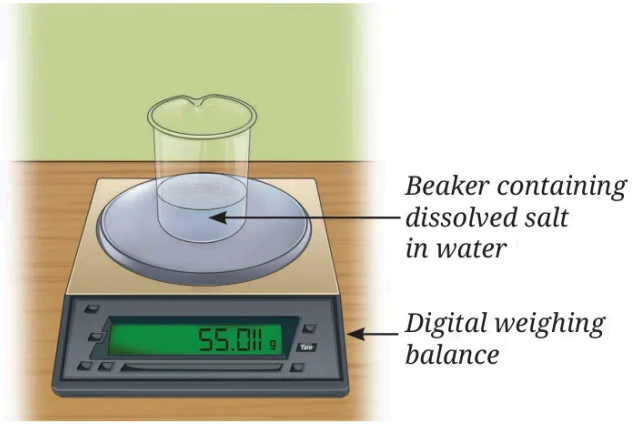


Fig. 9.1 (b) Salt solution: same reading

Observation

Mass of solution = mass of water + mass of salt. There is **no change in mass in a physical change**. (Tearing a sheet of paper into pieces also does not change its mass.)

Activity 9.2: Does mass change in a chemical change?

Vinegar reacts with baking soda (sodium hydrogencarbonate):

Vinegar + Baking soda → Carbon dioxide ↑ + Other substances

Set-up 1: open flask

Vinegar in a conical flask and baking soda in a balloon are weighed together (Fig. 9.2a). The baking soda is poured into the vinegar and the flask is weighed again.

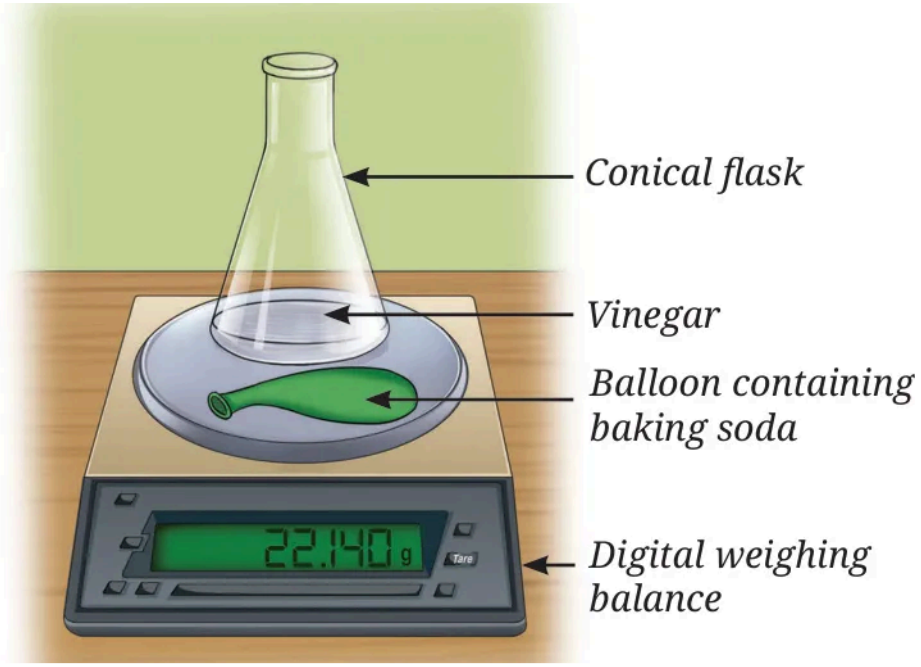


Fig. 9.2 (a) Weighing vinegar and baking soda before the reaction.

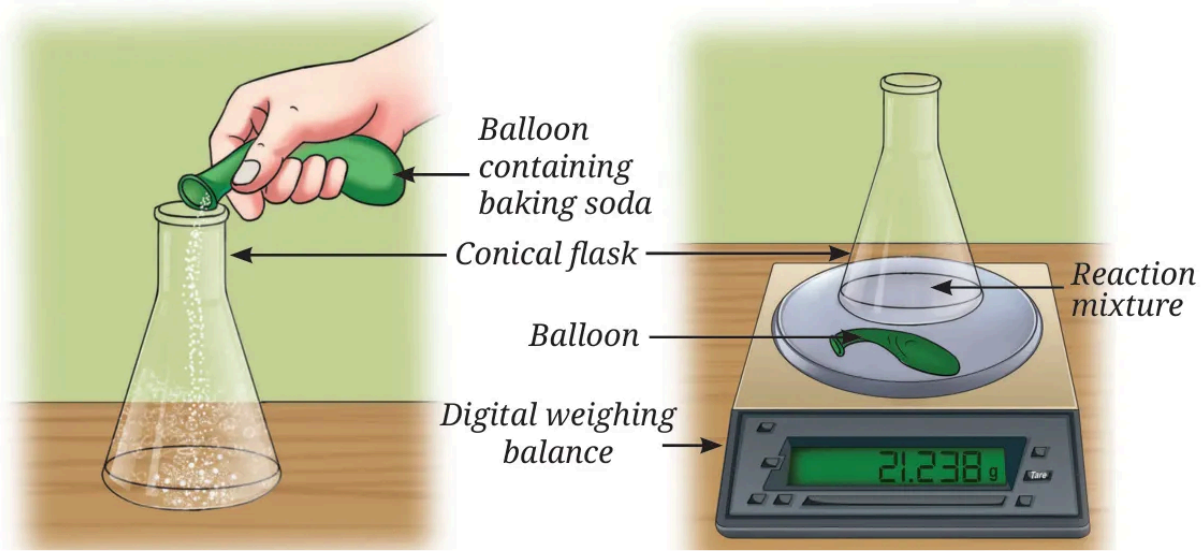


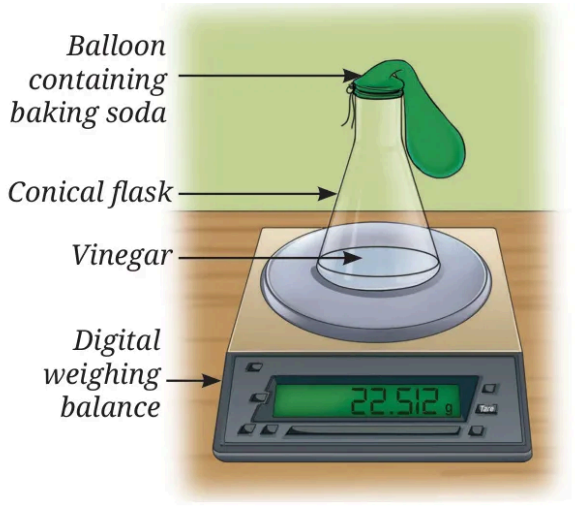
Fig. 9.2 (b), (c) Pouring baking soda into vinegar in an open flask, and the final reading.

Result: the reading goes DOWN

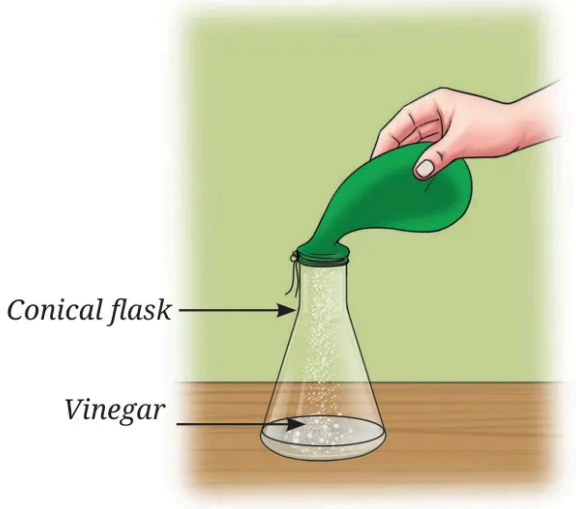
Brisk effervescence (fizzing) happens. The **carbon dioxide gas escapes** into the air from the open flask, so the final reading is less than the initial reading.

Set-up 2: balloon tied on the flask (closed system)

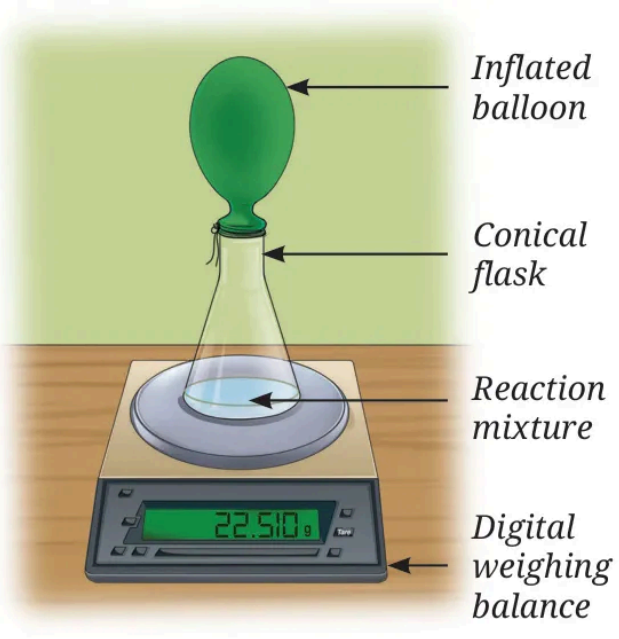
This time the balloon with baking soda is tied to the mouth of the flask. The whole set-up is weighed, then the balloon is lifted so the baking soda falls into the vinegar.



9.3 (a) Before



9.3 (b) Lift the balloon



9.3 (c) After: balloon inflates

Result: the reading stays the SAME

The gas is trapped in the balloon (it inflates). So **total mass of reactants = total mass of products**.

In set-up 1, mass seemed to be lost only because the gas escaped.

Law of Conservation of Mass

Matter (mass) can neither be created nor destroyed in a chemical reaction. Proposed by **Antoine Lavoisier** in **1789**.

Mass of reactants = Mass of products

Meet a scientist: Antoine Lavoisier

Known as the **Father of Modern Chemistry**. He wrote that "in every operation an equal quantity of matter exists both before and after the operation".

Activity 9.3: Verify the law (group activity)

Take about 10 mL of 1% sodium sulfate solution in flask A and 10 mL of 1% barium chloride solution in flask B. Weigh both together, mix B into A, and weigh both flasks again.

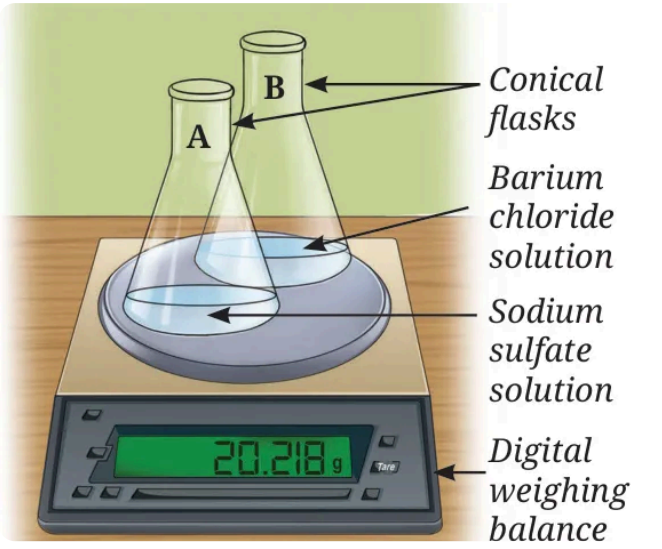


Fig. 9.4 (a) Reactants before mixing

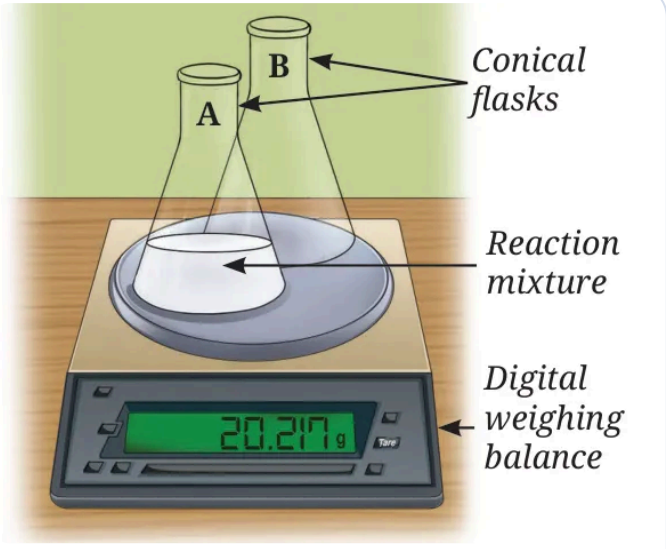


Fig. 9.4 (b) Products after mixing

Sodium sulfate + Barium chloride → Barium sulfate ↓ (white precipitate) + Sodium chloride

The reading does not change. No gas is formed, so this works even in an open flask.

📝 Exam tip: why keep both flasks on the balance?

A few drops always stick to the wall of the emptied flask. Keeping both flasks on the balance makes sure those drops are still counted. Also, a digital balance has an uncertainty of ±1 in its last digit, so a tiny change in the last digit is just experimental error.

🔬 **Think as a scientist**

Zinc + dilute hydrochloric acid → zinc chloride + hydrogen gas. To test whether mass is conserved, you must do it in a **closed** system (for example, tie a balloon over the flask) because hydrogen gas is formed and would escape.

Solved examples

Example 9.1

4.0 g calcium carbonate reacts with 2.92 g hydrochloric acid in a closed container, giving 1.76 g carbon dioxide, 0.72 g water and 4.44 g calcium chloride. Is the law obeyed?

Reactants: 4.0 + 2.92 = **6.92 g** · Products: 1.76 + 0.72 + 4.44 = **6.92 g**

Mass of reactants = mass of products, so the law is obeyed.

Example 9.2

12 g carbon + 32 g oxygen → 44 g carbon dioxide. How much CO_2 forms from 2.4 g carbon?

1 g carbon gives 44/12 g CO_2, so 2.4 g carbon gives (44/12) × 2.4

= 8.8 g of carbon dioxide

9.2 Law of Constant Proportions

Soon after Lavoisier, **Joseph Proust** showed that the elements in a compound are always present in a **fixed ratio by mass**, whatever the source of the compound.

Law of Constant (Definite) Proportions

In a compound, the elements are always combined in a fixed ratio by mass, no matter how or where the compound is obtained. Also called **Proust's Law**.

Example: water

Pure water from a river, borewell or ocean always has hydrogen : oxygen = **1 : 8** by mass.

So 9 g of water always breaks down into **1 g hydrogen + 8 g oxygen**.

Meet a scientist: Joseph Louis Proust

A French chemist. He showed that **copper carbonate** always contains copper, carbon and oxygen in the same proportion by mass, however it was prepared or wherever it was found.

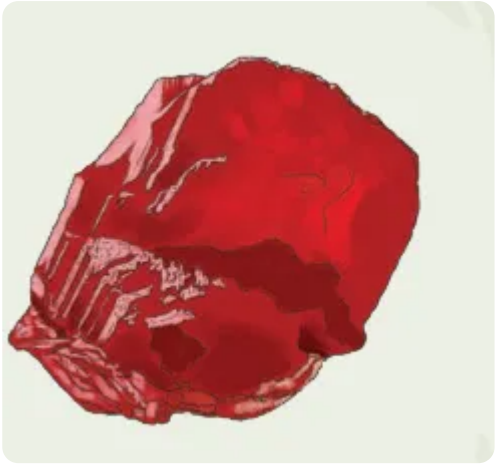


Fig. 9.5 Cinnabar (hingula)

🧵 Threads of curiosity: cinnabar

The red pigment **hingula** (cinnabar) gives **mercury 86.22%** and **sulfur 13.78%** by mass when heated. Grinding mercury and sulfur in this same ratio forms cinnabar again: a fixed ratio, just as Proust's law says.

Example 9.3

In NaCl, sodium : chlorine = 23 : 35.5 by mass. How much chlorine reacts with 46 g of sodium?

Chlorine = (35.5 ÷ 23) × 46 = 71 g

📝 Shortcut for ratio sums

If A : B = a : b and you have m g of A, then mass of B = **(b ÷ a) × m**. Always check that the units are grams on both sides.

9.3 Dalton's Atomic Theory

The two laws above formed the base of **John Dalton's** atomic theory (1808). A **postulate** is a basic assumption accepted as true without formal proof, from which further ideas are built.

Postulates of Dalton's atomic theory

1. All matter is made of very tiny particles called **atoms**, which take part in chemical reactions.
2. Atoms are **indivisible**; they cannot be created or destroyed in a chemical reaction. (explains conservation of mass)
3. Atoms of a given element are **identical** in mass and chemical properties.
4. Atoms of different elements have **different** masses and chemical properties.
5. Atoms combine in the ratio of **small whole numbers** to form compounds.
6. The relative number and kinds of atoms are **constant** in a given compound. (explains constant proportions)

Examples

Hydrogen and oxygen atoms combine to form water; the atoms are not destroyed. When magnesium burns in air, magnesium atoms join with oxygen atoms to form white magnesium oxide powder.

Remember from Chapter 8

We now know atoms **are** divisible: they contain electrons, protons and neutrons. But in a chemical reaction atoms are only **rearranged**, never created or destroyed, so Dalton's idea still explains the two laws.

🤔 What if...

...atoms could combine in any ratio? Then water from two sources could have different compositions and different properties, and no compound would have a fixed formula.

9.4 How do atoms combine?

Definition: molecule

A **molecule** is an electrically neutral particle made of more than one atom, that can exist independently and shows all the properties of that substance.

- **Molecule of an element:** same kind of atoms, e.g. H_2, Cl_2, O_2.
- **Molecule of a compound:** different kinds of atoms, e.g. HCl, H_2O.
- Some elements like **helium** already have a stable shell and exist only as single atoms.

Atoms with fewer than 8 valence electrons (fewer than 2 if K is the only shell) become stable in two ways:

Sharing electrons → covalent bond

Atoms share some of their valence electrons.

Transferring electrons → ionic bond

One atom gives electrons, the other takes them.

When atoms combine, the total energy becomes lower than that of the separate atoms, so the arrangement is more stable. The force that holds atoms together is called a **chemical bond**.

9.4.1 Covalent bond: bonding by sharing electrons

A. Molecules of elements

Hydrogen (H_2): H has 1 electron (K shell). It needs 1 more to fill the K shell (duplet), so two H atoms share one electron each.

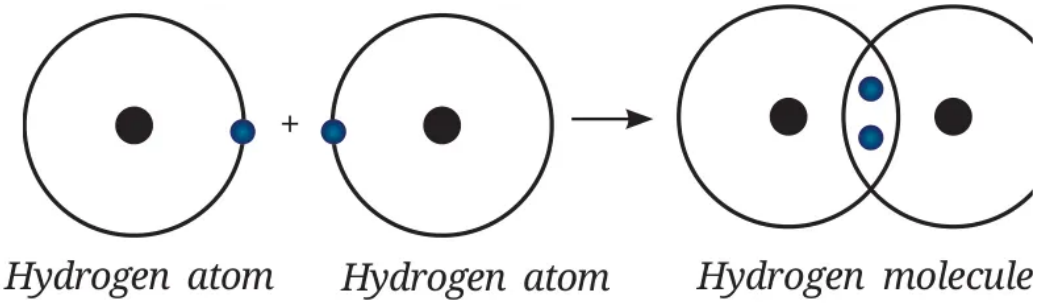


Fig. 9.6 Formation of a hydrogen molecule: H—H

Definitions

Covalent bond: the bond formed between atoms by a **shared pair of electrons**. The shared pair attracts both nuclei and holds them together.

Single bond: one shared pair (each atom gives 1 electron), drawn as one line, H—H.

Chlorine (Cl_2): Cl is 2, 8, 7. It needs 1 electron, so two Cl atoms share one electron each: a single bond, Cl—Cl.

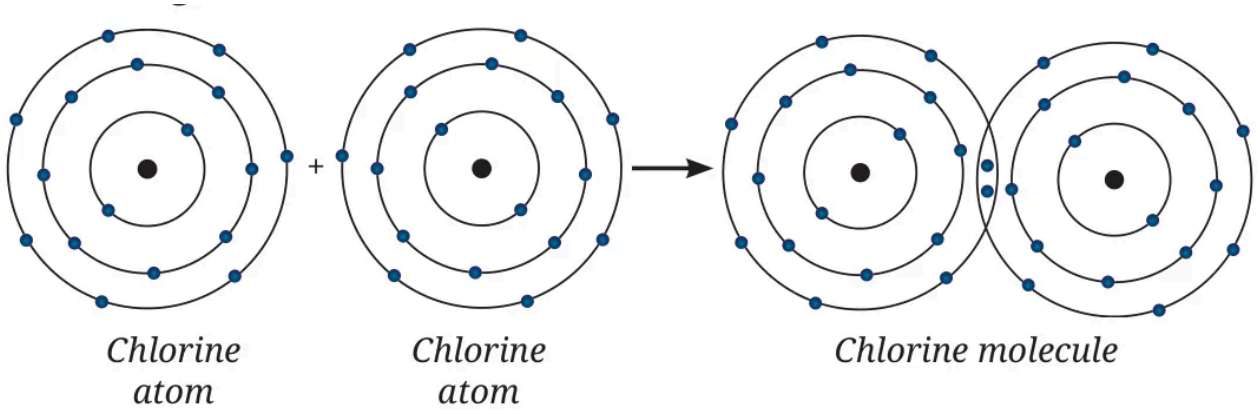


Fig. 9.7 Formation of a chlorine molecule: Cl—Cl

Oxygen (O_2): O is 2, 6. It needs 2 electrons, so two O atoms share **two** electrons each: a **double bond**, O=O.

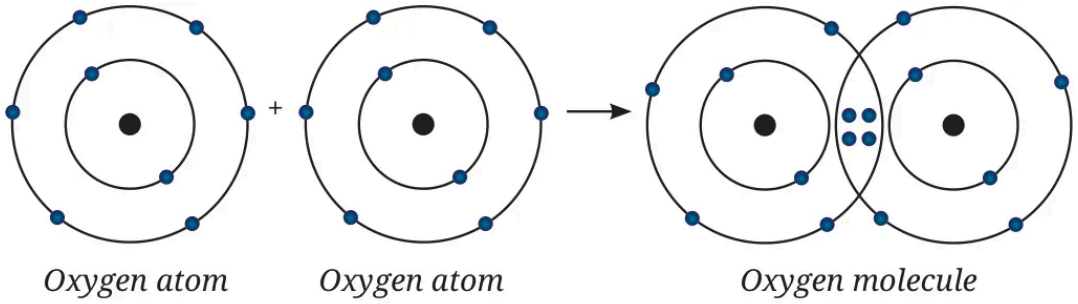


Fig. 9.8 Formation of an oxygen molecule: O=O (two shared pairs)

📝 Count the lines

Number of lines between two atoms = number of **shared pairs**. One pair: single bond (H—H). Two pairs: double bond (O=O). Three pairs: triple bond (N≡N in nitrogen).

B. Molecules of compounds

Steps: (1) write the electronic configurations; (2) count how many electrons each atom needs (octet, or duplet for H); (3) share that many electrons.

Hydrogen chloride (HCl): H (1) needs 1 and Cl (2, 8, 7) needs 1, so they share one pair: H—Cl. HCl is a **covalent compound**.

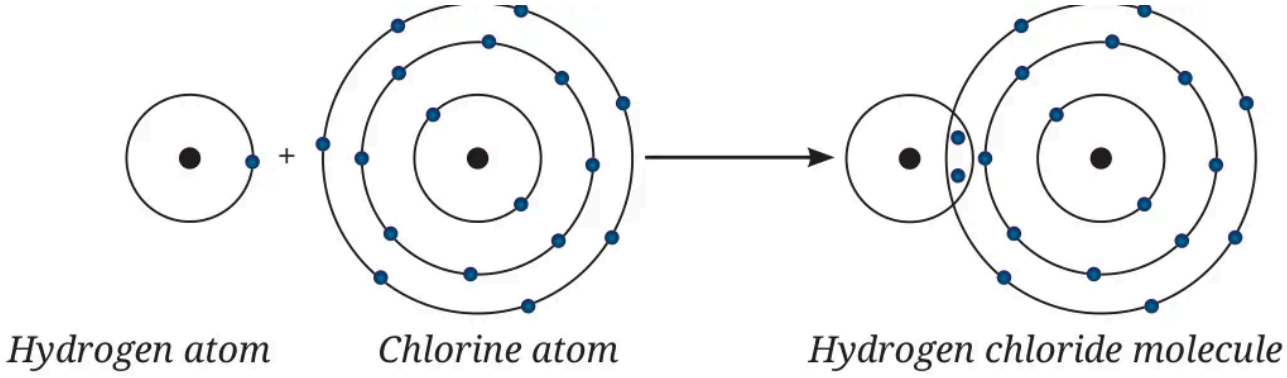


Fig. 9.9 Formation of a hydrogen chloride molecule: H—Cl

Water (H_2O): O needs 2 electrons but each H can share only 1. So **two** H atoms each share one electron with one O atom.

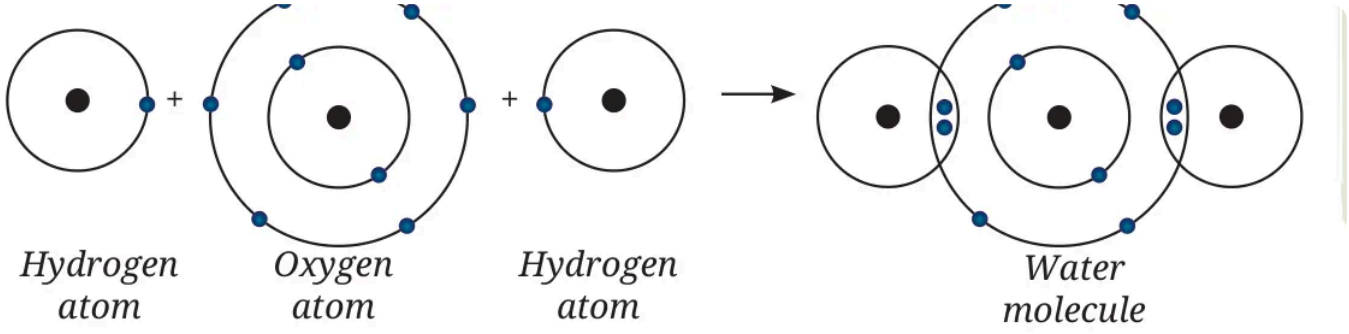


Fig. 9.10 Formation of a water molecule: H—O—H

🤝 Together we are strong

Just as atoms share electrons and become stable, sharing and caring among people builds strong relationships, a stronger community and a strong nation.

⚛️ Bridging science and society

Splitting or combining atomic nuclei releases huge **nuclear energy**, used to generate electricity, in medicine, research and space exploration. **Raja Ramanna**, often called the Father of the Indian Nuclear Programme, helped build India's peaceful nuclear energy programme.

C. Naming covalent compounds

- First element keeps its name; the second ends in **-ide**.
- Prefixes give the number of atoms: **mono** 1, **di** 2, **tri** 3, **tetra** 4, **penta** 5, **hexa** 6.
- **mono-** is dropped for the first element but used for the second.

- If the prefix ends in **o** or **a** and the element starts with a vowel, drop that vowel (monoxide, pentoxide). Prefixes ending in **i** keep it (dioxide, trioxide).
- No prefix before **hydrogen** when it comes first: H_2S is hydrogen sulfide.

Formula	Name	Note
CO	carbon monoxide	not "monooxide"
CO_2	carbon dioxide	not "monocarbon dioxide"
CS_2	carbon disulfide	two S atoms
PCl_3	phosphorus trichloride	three Cl atoms
SF_6	sulfur hexafluoride	six F atoms
N_2O_4	dinitrogen tetroxide	not "tetraoxide"
N_2O_5	dinitrogen pentoxide	
H_2O, NH_3	water, ammonia	common names are used

9.4.2 Ionic bond: transfer of electrons

Rule of thumb

Atoms with **fewer than 4** valence electrons usually **lose** them (metals: Na, Mg, Al...). Atoms with **more than 4** usually **gain** or share electrons (non-metals: O, S, Cl...).

Sodium chloride (NaCl)

Step 1: Sodium (Z = 11) is 2, 8, 1. It **loses** its one valence electron. It now has 11 protons and only 10 electrons, so it carries a +1 charge: the **sodium cation**, Na^+ (2, 8).

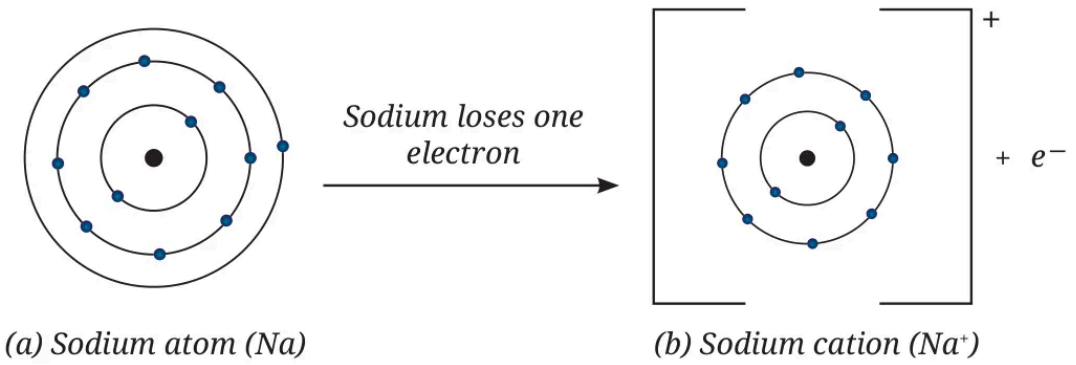


Fig. 9.11 Formation of a sodium cation: Na → $Na^+ + e^-$

Step 2: Chlorine (Z = 17) is 2, 8, 7. It **gains** one electron: 17 protons, 18 electrons, charge –1. This is the **chloride anion**, Cl^- (2, 8, 8).

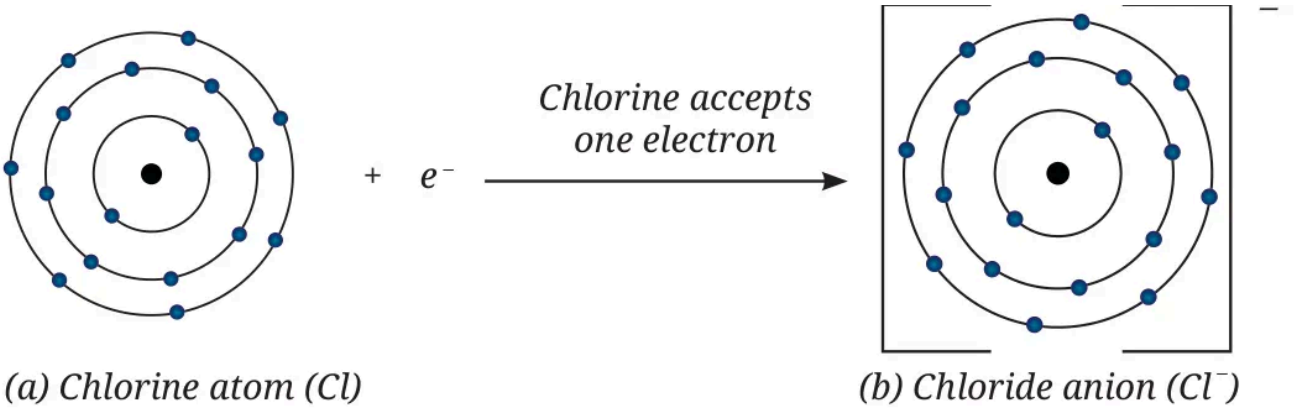


Fig. 9.12 Formation of a chloride anion: Cl + e^- → Cl^-

Step 3: The oppositely charged ions attract each other.

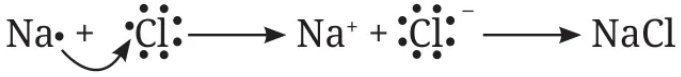


Fig. 9.13 Formation of NaCl by transfer of an electron

Definitions

Cation: positive ion (atom lost electrons). **Anion**: negative ion (atom gained electrons). Together they are called **ions**.

Ionic bond: the electrostatic force of attraction between oppositely charged ions that holds them together.

🧵 Threads of curiosity: crystals

Ionic compounds do not exist as single units. In a NaCl crystal each Na^+ is surrounded by six Cl^- ions and each Cl^- by six Na^+ ions, in a repeating 3-D pattern called a **crystal structure**. Drawn with dots for ions, it is called a **crystal lattice**.

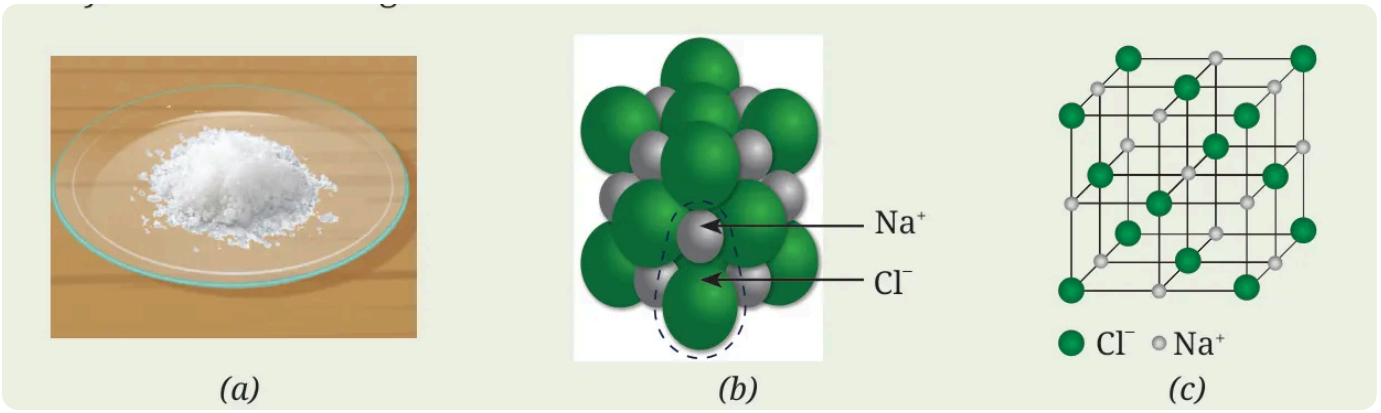


Fig. 9.14 Sodium chloride: (a) crystals, (b) crystal structure, (c) crystal lattice

Sulfur (2, 8, 6) gains **two** electrons to complete its octet and becomes the sulfide ion, S^{2-}.

A. Naming ionic compounds

- Write the **cation first**, then the anion: sodium chloride, calcium oxide, magnesium sulfide.
- Simple anions end in **-ide** (chloride, oxide, sulfide).
- Metals usually form cations; non-metals form anions.
- **Polyatomic ions** are made of two or more atoms (OH^-, SO_4^{2-}). Their names usually do **not** end in -ide (hydroxide is an exception).

Table 9.1 (a) Common monoatomic ions

Ion	Formula	Valency
Sodium	Na^+	1
Lithium	Li^+	1
Potassium	K^+	1
Silver	Ag^+	1
Copper (cuprous)	Cu^+	1
Calcium	Ca^{2+}	2
Barium	Ba^{2+}	2
Magnesium	Mg^{2+}	2
Zinc	Zn^{2+}	2
Iron (ferrous)	Fe^{2+}	2
Copper (cupric)	Cu^{2+}	2

Iron (ferric)	Fe^{3+}	3
Aluminium	Al^{3+}	3
Fluoride	F^{-}	1
Chloride	Cl^{-}	1
Bromide	Br^{-}	1
Iodide	I^{-}	1
Oxide	O^{2-}	2
Sulfide	S^{2-}	2

Table 9.1 (b) Common polyatomic ions

Ion	Formula	Valency
Hydroxide	OH^{-}	1
Nitrate	NO_3^{-}	1
Hydrogencarbonate	HCO_3^{-}	1
Ammonium	NH_4^{+}	1
Carbonate	CO_3^{2-}	2
Sulfate	SO_4^{2-}	2

📝 **Memory tip**

-ous = lower charge (ferrous Fe^{2+}, cuprous Cu^{+}). **-ic** = higher charge (ferric Fe^{3+}, cupric Cu^{2+}). Ammonium NH_4^{+} is the only **positive** polyatomic ion in the table.

9.5 Writing chemical formulae

The quick way is the **criss-cross method**: swap the valencies (or charges) of the two parts and write them as subscripts.

9.5.1 Covalent compounds

1. Write the symbols of the elements.
2. Write their valencies under them.

3. Cross over the valencies and write them as subscripts. A subscript of 1 is not written.

Symbol	H	Cl
Valency	1	1
Formula	**HCl**	

Symbol	H	S
Valency	1	2
Formula	**H_2S**	

Symbol	C	Cl
Valency	4	1
Formula	**CCl_4**	

9.5.2 Ionic compounds

1. Write the cation first, then the anion.
2. Write the charges under the symbols.
3. Cross over only the **numbers** of the charges.
4. Divide the subscripts by any common factor to get the simplest ratio (2 and 4 become 1 and 2).

Symbol	Ca	Cl
Charge	2+	1–
Formula	**$CaCl_2$**	

Symbol	Al	O
Charge	3+	2–
Formula	**Al_2O_3**	

Symbol	Mg	O
Charge	2+	2–
Formula	Mg_2O_2 → **MgO**	

In $CaCl_2$ there are two Cl^- ions for each Ca^{2+} ion, so the charges balance (+2 –2 = 0). The charges are **not** written in the final formula.

With polyatomic ions

Symbol	Ca	CO_3
Charge	2+	2–
Formula	$Ca_2(CO_3)_2$ → **$CaCO_3$**	

Symbol	Mg	OH
Charge	2+	1–
Formula	**$Mg(OH)_2$**	

Symbol	Al	OH
Charge	3+	1–
Formula	**$Al(OH)_3$**	

Symbol	Al	SO_4
Charge	3+	2–
Formula	**$Al_2(SO_4)_3$**	

Common mistake: brackets

Use brackets when there are **two or more** of the same polyatomic ion: $Al(OH)_3$, **not** $AlOH_3$. No brackets are needed for just one: $CaCO_3$, NaOH.

Properties of ionic and covalent compounds

Activity 9.4

Take camphor, sodium chloride, copper sulfate, sugar and naphthalene. Test (A) whether each dissolves in water, kerosene and petrol, and (B) whether it conducts electricity as a solid and when dissolved in water, using two electrodes, a 9 V battery and a bulb.

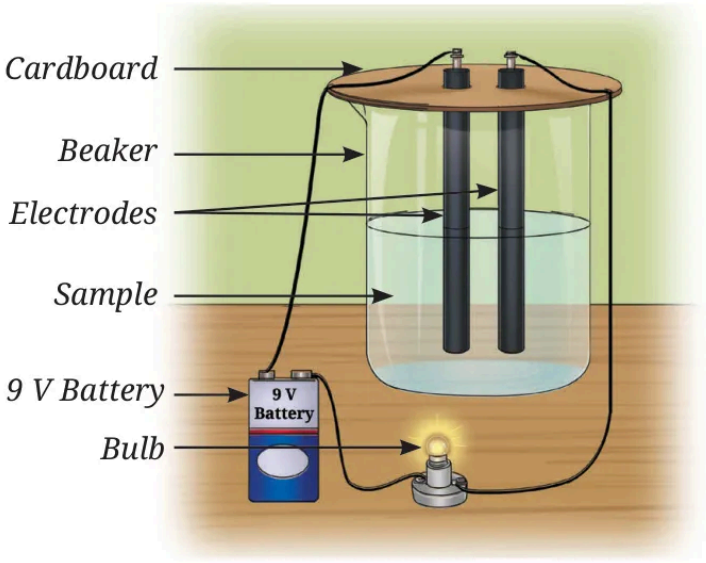


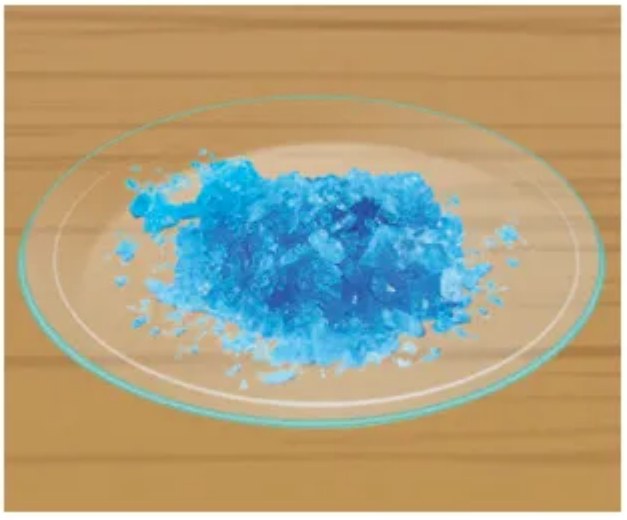
Fig. 9.15 Set-up to test the electrical conductivity of a solution

⚠ Safety first

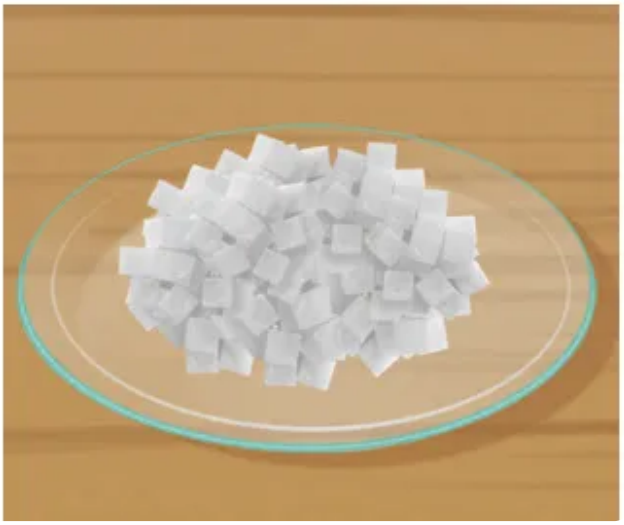
Do not touch the electrodes when connected; use only a low-voltage battery. Petrol and kerosene catch fire easily; keep them away from flames.



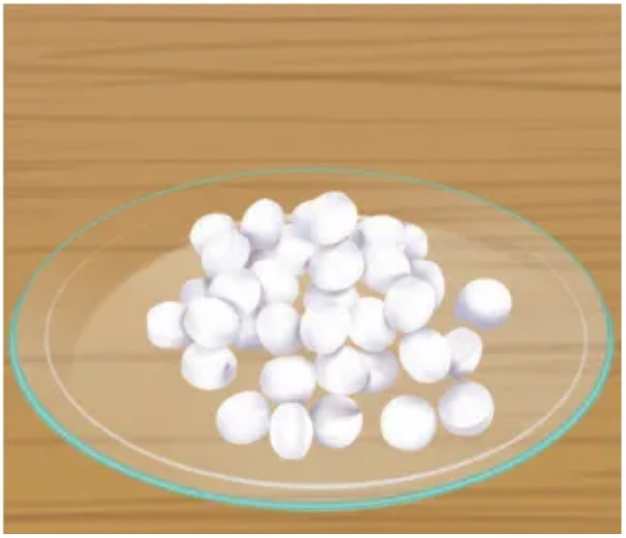
9.16 (a) Sodium chloride (ionic)



9.16 (b) Copper sulfate (ionic)



9.17 (a) Camphor (covalent)



9.17 (b) Naphthalene (covalent)

Property	Ionic compounds	Covalent compounds
Formed by	Transfer of electrons	Sharing of electrons
Solubility	Generally soluble in water; insoluble in kerosene, petrol	Mostly insoluble in water (sugar is an exception); soluble in kerosene, petrol
Conduction as solid	No (ions fixed in place)	No
Conduction in water or molten state	Yes (ions free to move)	No (no ions)
Melting and boiling points	High (strong attraction between ions)	Usually low
Examples	NaCl, $CuSO_4$	Sugar, camphor, naphthalene

📝 **Why is sugar solution not a conductor?**

Sugar dissolves in water, but as whole **molecules**, not ions. To conduct electricity you need **free-moving ions**.

9.7–9.8

Molecular mass and formula unit mass

Molecular mass = sum of atomic masses of all atoms in a molecule

Used for covalent compounds. Unit: u (atomic mass unit).

Formula unit mass = sum of atomic masses of all atoms in a formula unit

Used for ionic compounds, which form 3-D crystals, not molecules. A **formula unit** is the simplest whole-number ratio of ions.

Atomic masses to remember

H = 1 u · C = 12 u · N = 14 u · O = 16 u · Na = 23 u · Mg = 24 u · S = 32 u · Cl = 35.5 u · K = 39 u · Ca = 40 u

Example 9.4: H_2O

(1 u × 2) + (16 u × 1)

= 18 u

Example 9.5: CO_2

(12 u × 1) + (16 u × 2)

= 44 u

Example 9.6: Na_2O

(23 u × 2) + (16 u × 1)

= 62 u

Example 9.7: $Ca(NO_3)_2$

40 + {14 + (16 × 3)} × 2 = 40 + 62 × 2

= 164 u

📝 **Brackets first**

For $Ca(NO_3)_2$, first find the mass of one NO_3 (62 u), then multiply by 2. Forgetting the bracket is the most common mistake.

Recap

At a glance

- **Law of Conservation of Mass** (Lavoisier, 1789): mass can neither be created nor destroyed in a chemical reaction.

- **Law of Definite Proportions** (Proust): a compound always contains the same elements in a fixed ratio by mass (water H : O = 1 : 8).
- **Dalton's Atomic Theory** (1808) explained both laws: atoms are only rearranged in reactions and combine in simple whole-number ratios.
- A **molecule** is a neutral particle of more than one atom that can exist independently.
- Atoms combine to become stable; the force holding them is a **chemical bond**.
- **Covalent bond**: sharing of electrons (single H—H, double O=O).
- **Ionic bond**: transfer of electrons forming cations and anions held by electrostatic attraction.
- The formula of a covalent compound gives the actual number of atoms; the formula of an ionic compound gives the **simplest whole-number ratio**.
- **Molecular mass**: sum of atomic masses in a molecule. **Formula unit mass**: sum of atomic masses in a formula unit of an ionic compound.

🔭 Explore more

- Build molecules yourself: PhET Build a Molecule simulation
- Design an experiment to show that water from any source has H and O in the same ratio.
- Make a card game with cation and anion cards and match them to form compounds.

✨ The quest continues

Are there any chemical changes that do not obey the Law of Conservation of Mass?

Quiz Test yourself

Tap an option to see the right answer and the reason.

1. The Law of Conservation of Mass was proposed by...

John Dalton	Joseph Proust
Antoine Lavoisier	J.J. Thomson

Lavoisier, 1789. Proust gave the law of constant proportions.

2. In set-up 1 of Activity 9.2 (open flask) the final reading is less because...

- mass is destroyed
- **CO_2 gas escapes**
- vinegar evaporates
- the balance is faulty

The gas formed leaves the open flask; in the closed set-up the reading stays the same.

3. Hydrogen and oxygen in water are in the mass ratio...

- 2 : 1
- 1 : 16
- **1 : 8**
- 8 : 1

9 g water → 1 g hydrogen + 8 g oxygen.

4. How many shared pairs of electrons are there in O_2?

- 1
- **2**
- 3
- 4

Each O needs 2 electrons, so they form a double bond, O=O.

5. A sodium ion Na^+ has...

- 11 p, 11 e
- **11 p, 10 e**
- 10 p, 11 e
- 12 p, 10 e

It loses one electron but keeps all 11 protons.

6. The correct name of N_2O_5 is...

- nitrogen pentaoxide
- dinitrogen pentaoxide
- **dinitrogen pentoxide**
- nitrogen oxide

"penta" ends in a and oxide starts with a vowel, so the a is dropped.

7. The formula of aluminium sulfate is...

- $AlSO_4$
- $Al_3(SO_4)_2$
- **$Al_2(SO_4)_3$**
- Al_2SO_4

Al^{3+} and SO_4^{2-}: criss-cross 3 and 2.

8. A solid that conducts electricity only when dissolved in water has...

- **ionic bonds**
- covalent bonds
- no bonds
- metallic bonds only

Its ions are fixed in the solid and become free in water.

9. The molecular mass of CH_4 is...

- 13 u
- **16 u**
- 18 u
- 20 u

12 + (1 × 4) = 16 u.

10. The formula unit mass of $Mg(OH)_2$ is...

41 u	48 u
58 u	74 u

24 + 2 × (16 + 1) = 58 u.

Answers to all questions

Tap a question to see the answer. Use **Print or save as PDF** at the top to get all answers open.

Pause and Ponder

1. A student burns 10 g of ethanol in an open beaker. No residue is left. Does this mean the Law of Conservation of Mass is violated?

No. Ethanol burns with oxygen of the air to form **carbon dioxide and water vapour**, which are gases and escape from the open beaker. Mass of ethanol + oxygen used = mass of CO_2 + water vapour formed. If the reaction is done in a closed container, the total mass stays the same.

2. When 20 g of hydrogen reacts completely with 160 g of oxygen, how much water is formed?

Mass of water = 20 g + 160 g = **180 g** (mass of reactants = mass of products). Note that 20 : 160 = 1 : 8, the fixed ratio for water.

3. A compound has 40% sulfur and 60% oxygen by mass. A sample has 20 g sulfur. How much oxygen is present?

S : O = 40 : 60 = 2 : 3. Oxygen = (60 ÷ 40) × 20 = **30 g**.

4. In CO, carbon : oxygen = 3 : 4. How much oxygen combines with 9 g of carbon?

Oxygen = (4 ÷ 3) × 9 = **12 g**.

5. The Law of Definite Proportions holds for compounds but not for mixtures. Give reason.

In a **compound**, elements are chemically combined in a fixed ratio by mass. In a **mixture**, the components are only physically mixed and can be present in **any** ratio (for example, salt solution can be weak or strong), so there is no fixed proportion.

6. Students X and Y made an oxide of copper with copper : oxygen = 4 : 1 and 8 : 2. Do their results justify the law?

Yes. 8 : 2 simplifies to **4 : 1**, the same ratio as X. The same compound has the same fixed mass ratio, as the Law of Constant Proportions says.

7. Assertion (A): 2 g hydrogen combines with 16 g oxygen to form 18 g water. Reason (R): According to Dalton, atoms combine in a simple whole number ratio to form compounds.

(ii) Both A and R are true, but R is not the correct explanation of A. The 18 g of water is explained by the **Law of Conservation of Mass** (2 g + 16 g = 18 g, atoms are not created or destroyed). R is a true postulate, but it is about the ratio in which atoms combine, not about the total mass.

8. Nitrogen has five valence electrons. Draw the structure of the nitrogen molecule (N_2).

N is 2, 5 and needs 3 more electrons. Each N atom shares **three** electrons with the other, making three shared pairs, a **triple bond**: **N≡N**. Each N also keeps one lone pair, so both get an octet. In the drawing, show the two atoms' outer shells overlapping with 6 electrons (3 pairs) in the overlap.

9. The atomic number of fluorine is 9. Explain the formation of F_2.

F is 2, 7 and needs 1 electron. Two F atoms share one electron each, forming one shared pair: a single covalent bond, **F—F**. Both atoms now have 8 electrons in the outer shell (just like Cl_2 in Fig. 9.7).

10. Show the formation of (i) CO_2, (ii) H_2S, (iii) NH_3.

(i) **CO_2**: C (2, 4) needs 4; each O (2, 6) needs 2. C shares two pairs with each O: **O=C=O** (two double bonds).
(ii) **H_2S**: S (2, 8, 6) needs 2; each H needs 1. S shares one pair with each of two H atoms: **H—S—H** (like water).
(iii) **NH_3**: N (2, 5) needs 3. N shares one pair with each of three H atoms; N keeps one lone pair. Three single bonds N—H.

11. Neon (atomic number 10) neither transfers nor shares its valence electrons. Explain.

Neon is **2, 8**. Its outermost shell already has a complete octet, so it is already stable and has no need to lose, gain or share electrons. That is why neon exists as single atoms.

12. What kind of ion will oxygen form?

O is 2, 6. It gains 2 electrons to complete its octet and forms the **oxide anion, O^{2-}** (a negative ion with charge 2–).

13. Fill in the blanks: magnesium gives two electrons to become Mg^{2+}; chlorine takes one electron to become ___. ___ ion of magnesium and ___ ions of chlorine combine to give magnesium chloride.

Cl^-; **one**; **two**. $Mg^{2+} + 2Cl^- \rightarrow MgCl_2$.

14. Show the formation of cations of K and Ca, and of their chlorides.

K (2, 8, 8, 1) loses 1 electron → **K^+** (2, 8, 8).
Ca (2, 8, 8, 2) loses 2 electrons → **Ca^{2+}** (2, 8, 8).
Cl (2, 8, 7) gains 1 electron → Cl^- (2, 8, 8).
$K^+ + Cl^- \rightarrow$ **KCl** · $Ca^{2+} + 2Cl^- \rightarrow$ **$CaCl_2$** (Ca gives one electron to each of two Cl atoms).

15. Illustrate how sodium sulfide (Na_2S) is formed.

Na (2, 8, 1) loses 1 electron; S (2, 8, 6) needs 2. So **two** sodium atoms each give one electron to one sulfur atom: $2Na \rightarrow 2Na^+ + 2e^-$; $S + 2e^- \rightarrow S^{2-}$. Then $2Na^+ + S^{2-} \rightarrow$ **Na_2S**.

16. Name: (i) CO_2 (ii) NO_2 (iii) SF_6 (iv) PCl_3

(i) carbon dioxide (ii) nitrogen dioxide (iii) sulfur hexafluoride (iv) phosphorus trichloride

17. Write the formula: (i) sodium hydrogencarbonate (ii) sulfur dioxide (iii) ferric chloride (iv) cuprous oxide

(i) **$NaHCO_3$** (ii) **SO_2** (iii) **$FeCl_3$** (Fe^{3+}, Cl^-) (iv) **Cu_2O** (Cu^+, O^{2-})

18. Write formulae for compounds of: (i) Fe^{3+} and OH^- (ii) K^+ and CO_3^{2-}

(i) **$Fe(OH)_3$** (ii) **K_2CO_3**

19. **What type of bond is in a solid that does not conduct as a solid but conducts when dissolved in water?**

Ionic bond. The ions are fixed in the solid crystal but become free to move in water.

20. **Metal M has two electrons in its valence shell (M shell) and reacts with oxygen. Predict (i) formula (ii) type of bond (iii) conductivity of its aqueous solution.**

M is 2, 8, 2, which is **magnesium**. It forms M^{2+}; oxygen forms O^{2-}.
(i) Formula: **MO** (MgO) (ii) **Ionic** bond (iii) Its aqueous solution **conducts electricity**, because the dissolved part gives free ions (weakly, since it is only slightly soluble).

21. **Find the molecular mass of nitric acid (HNO_3).**

$1 + 14 + (16 \times 3) =$ **63 u**

22. **Find the molecular mass of methane (CH_4).**

$12 + (1 \times 4) =$ **16 u**

23. **Find the formula unit mass of KCl.**

$39 + 35.5 =$ **74.5 u**

24. **Find the formula unit mass of $Mg(OH)_2$.**

$24 + 2 \times (16 + 1) = 24 + 34 =$ **58 u**

Revise, Reflect, Refine

1. **Element A has one electron in its third shell; element B has six electrons in its second shell. (i)–(vi)**

A = 2, 8, 1 (sodium); B = 2, 6 (oxygen).
(i) A tends to **give 1** electron. (ii) A forms a **cation, A^+**. (iii) B tends to **take 2** electrons. (iv) B forms an **anion, B^{2-}**. (v) An **ionic** bond. (vi) Formula **A_2B** (like Na_2O).

2. Element X has six electrons in its outer shell and forms a diatomic molecule. (i)–(iv)

(i) X needs 2 more electrons for an octet. Two X atoms share 2 electrons each, so both become stable as an X_2 molecule.
(ii) A **covalent** bond (a double bond).
(iii) **X=X**, like O_2 in Fig. 9.8: two shared pairs between the atoms.
(iv) Y is 2, 2 (two electrons in the second shell, like beryllium). Y gives its 2 valence electrons to X, forming Y^{2+} and X^{2-}. The compound is **YX**, drawn as $[Y]^{2+}$ $[X]^{2-}$ with X having 8 electrons in its outer shell.

3. Total positive charge 6+ and total negative charge 6–: which combination is correct?

(iii) 2 Fe^{3+} and 3 O^{2-}: 2 × 3 = 6+ and 3 × 2 = 6–.
(i) 3 Cl^- gives only 3–. (ii) 1 PO_4^{3-} gives only 3–. (iv) 2 SO_4^{2-} gives only 4–.

4. Choose the correct statement(s) and correct the false ones.

(i) **False.** Elements are made of one kind of atom (as single atoms or molecules like O_2); compounds are made of molecules or ions containing atoms of different elements.
(ii) **False.** A molecule of a compound is always made of atoms of two or more **different** kinds.
(iii) **False.** One molecule of nitrogen gas (N_2) contains **two** nitrogen atoms.
(iv) **True.**

5. Write formulae: (i) aluminium nitrate (ii) calcium oxide (iii) ferric oxide

(i) **$Al(NO_3)_3$** (ii) **CaO** (iii) **Fe_2O_3**

6. Formulae from: (i) Ca^{2+}, Br^- (ii) Al^{3+}, CO_3^{2-} (iii) K^+, SO_4^{2-} (iv) NH_4^+, Cl^-

(i) **$CaBr_2$** (ii) **$Al_2(CO_3)_3$** (iii) **K_2SO_4** (iv) **NH_4Cl**

7. Which diagram in Fig. 9.18 correctly represents the Cl^- ion?

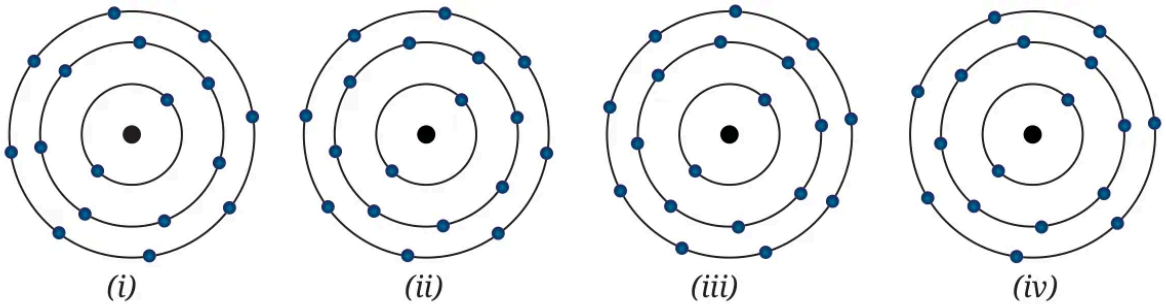


Fig. 9.18 Count the electrons in each shell.

Cl^- has 17 + 1 = 18 electrons, arranged **2, 8, 8**. Counting the dots: (i) 2, 7, 8 (ii) **2, 8, 8** (iii) 2, 8, 9 (iv) 2, 8, 7 (that is a neutral Cl atom). So the answer is **(ii)**.

8. Formula unit mass of (i) NH_4NO_3 (ii) H_3PO_4 (iii) $NaHCO_3$

(i) 14 + 4 + 14 + 48 = **80 u**

(ii) 3 + 31 + 64 = **98 u** (P = 31 u)

(iii) 23 + 1 + 12 + 48 = **84 u**

9. Formulae of compounds formed by: (i) Mg and N (ii) Li and N (iii) Na and S (iv) Al and O

(i) $Mg^{2+}, N^{3-} \rightarrow$ **Mg_3N_2** (ii) $Li^+, N^{3-} \rightarrow$ **Li_3N** (iii) **Na_2S** (iv) **Al_2O_3**

10. Complete Table 9.3.

	NO_3^-	SO_4^{2-}	PO_4^{3-}
NH_4^+	NH_4NO_3	$(NH_4)_2SO_4$	$(NH_4)_3PO_4$
Li^+	$LiNO_3$	Li_2SO_4	Li_3PO_4
Al^{3+}	$Al(NO_3)_3$	$Al_2(SO_4)_3$	$AlPO_4$
Cu^{2+}	$Cu(NO_3)_2$	$CuSO_4$	$Cu_3(PO_4)_2$

11. **5.3 g sodium carbonate + 6.0 g acetic acid → 2.2 g CO_2 + 0.9 g water + 8.2 g sodium acetate. Verify the law.**

Reactants: 5.3 + 6.0 = **11.3 g**. Products: 2.2 + 0.9 + 8.2 = **11.3 g**. They are equal, so the Law of Conservation of Mass is **valid**.

12. **A species has 11 protons, 12 neutrons and 10 electrons. (i)–(iv)**

(i) Atomic number = 11; mass number = 11 + 12 = **23**.
(ii) A **cation**: 11 positive and only 10 negative charges, so net charge +1.
(iii) **2, 8**
(iv) **Sodium ion, Na^+**

13. **A: 2, 8, 5 and B: 2, 8, 7. (i)–(iii)**

A is phosphorus, B is chlorine.
(i) **B** is more reactive: it needs only one electron to complete its octet, while A needs three.
(ii) Both are non-metals that need electrons, so neither gives electrons away. They **share** electrons and form **covalent** bonds: A shares one electron with each of three B atoms, and each B shares one electron with A.
(iii) **AB_3** (like PCl_3).

14. **A: Copper sulfate conducts in the molten state but not in the solid state. R: Ions are fixed in the molten state but free in the solid state.**

(iii) A is true, but R is false. It is the other way round: ions are **fixed** in the solid and **free to move** in the molten state.

15. **^{27}Al, $^{80}Br^-$ and $^{201}Hg^{2+}$ have 13, 35 and 80 protons. How many electrons and neutrons?**

Species	Electrons	Neutrons
^{27}Al	13 (neutral)	27 – 13 = 14
$^{80}Br^-$	35 + 1 = 36	80 – 35 = 45
$^{201}Hg^{2+}$	80 – 2 = 78	201 – 80 = 121

What if...?

What if we could see atoms directly? How would it help scientists, and what challenges would it cause?

Scientists could watch bonds forming and breaking, and design new medicines and materials more easily. But atoms are always moving and are about 10^{-10} m across, far smaller than the wavelength of visible light, so any tool that "sees" them can also disturb them. The amount of information would also be enormous. (Open-ended: any well-reasoned answer is fine.)

MT Classes, S.K. Nagar, Motihari

Maths and Science coaching for Classes 8 to 10 (CBSE)

Notes based on NCERT Exploration (Grade 9 Science), Chapter 9: Atomic Foundations of Matter. Diagrams are taken from the NCERT textbook chapter.